

PART 2 - Before Adding More BESS, Should We Add Wind?

If Solar + BESS can increase renewable-energy utilisation, the obvious question is: why not simply keep increasing battery capacity until most grid consumption is replaced?

Because storage is not generation and it is costly.

Every additional unit of energy routed through a battery carries battery capex, round-trip efficiency loss, degradation, augmentation / replacement requirements, financing cost and utilisation risk.

There is another way to reduce the mismatch between renewable generation and industrial consumption: improve the generation profile itself.

Solar and Wind can be naturally complementary. Solar generation is concentrated during daylight hours, while Wind can generate across daytime, evening and night periods and has a different seasonal profile. A well-designed Wind-Solar portfolio can therefore align more closely with the consumer's load curve before storage is added.

Illustrative comparison: Wind vs Wind + Solar + BESS

Parameter	Wind CAPEX	Wind Group Captive	Wind + Solar + BESS CAPEX	Wind + Solar + BESS Group Captive
Configuration	4.25 MW Wind	4.25 MW Wind	3.15 MW Wind + 5.3 MWp Solar + 15 MWh BESS	3.15 MW Wind + 5.3 MWp Solar + 15 MWh BESS
RE / Grid replacement	61.1%	61.1%	81.4%	81.4%
Total landed RE cost	₹4.83/kWh	₹6.18/kWh	₹5.91/kWh	₹7.32/kWh
Balance grid consumption	38.9%	38.9%	18.6%	18.6%
Weighted average saving (on total consumption)	₹2.97/kWh	₹2.15/kWh	₹2.89/kWh	₹1.75/kWh

Illustrative scenario - key assumptions

- Same indicative Maharashtra industrial load profile used in Part 1.
- Standalone Wind configuration: 4.25 MW. Hybrid configuration: 3.15 MW Wind + 5.3 MWp Solar + 15 MWh BESS.
- BESS cost assumption: approximately ₹5.50/kWh of energy delivered through storage.
- Transmission charge assumption: ₹1.02/kWh.
- Applicable losses, Electricity Duty, ToD tariffs and banking treatment are as used in the underlying Maharashtra model.
- Indicative lapse / unutilised renewable energy: approximately 5.1% for Wind and 5.0% for Wind + Solar + BESS.

- All values are illustrative and should be recalculated using the customer's actual interval load data and final project / regulatory assumptions.

The hybrid configuration increases renewable replacement to 81.4% - but maximum renewable replacement is not the same as maximum savings.

In the illustrative CAPEX case, standalone Wind produces slightly higher weighted average savings (₹2.97/kWh) than Wind + Solar + BESS (₹2.89/kWh), even though the hybrid configuration reduces grid dependence much further. This is exactly why storage should be optimised rather than maximised.

A balanced Wind + Solar portfolio can increase renewable replacement, reduce excess midday Solar, provide renewable energy during non-solar hours, reduce BESS cycling requirements and potentially reduce the storage capacity required to achieve the same customer outcome.

**Optimise Generation → Optimise Consumption → Identify the Residual ToD Gap →
Size BESS**

BESS should therefore be the final optimisation layer - not the starting point. The objective is to achieve the required renewable-energy replacement and savings with the minimum technically and commercially viable storage requirement.

For consumers targeting high renewable-energy replacement, the relevant question is increasingly not "Solar versus BESS". It is:

"What combination of Wind + Solar + Storage produces the lowest landed cost for the customer's actual load profile?"

In Part 3, we address the harder issue: even where BESS makes technical and commercial sense, is Maharashtra's Open Access framework ready to account for it consistently?

